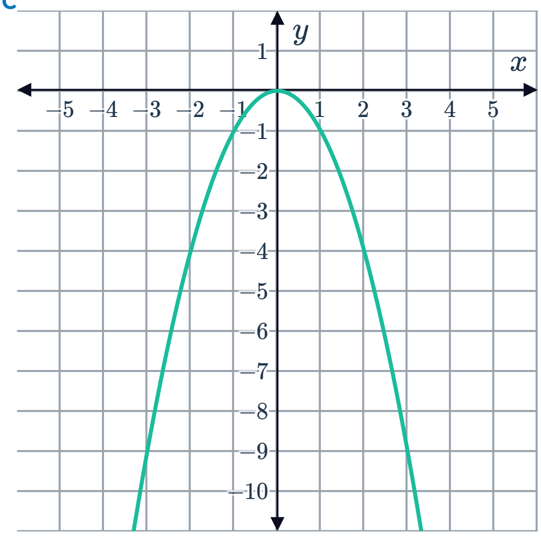
Finding the equation of a parabola

1

a $a = -1$

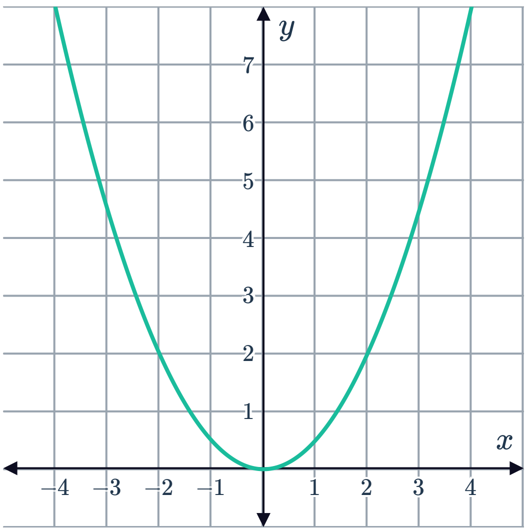
b $(0, 0)$

c



2

a



b $y = \frac{1}{2}x^2$

3

a No

b Yes

4

a $(3, 5)$

b $y = (x - 3)^2 + 5$

5

a $x = -7$

b $y = -\frac{3}{7}(x - 1)(x + 7)$

c $\left(-3, \frac{48}{7}\right)$

7

a $x=-2$

b $y=(x+2)^2-9$

8 $y=x^2+6x-40$

9

a x^2-b^2

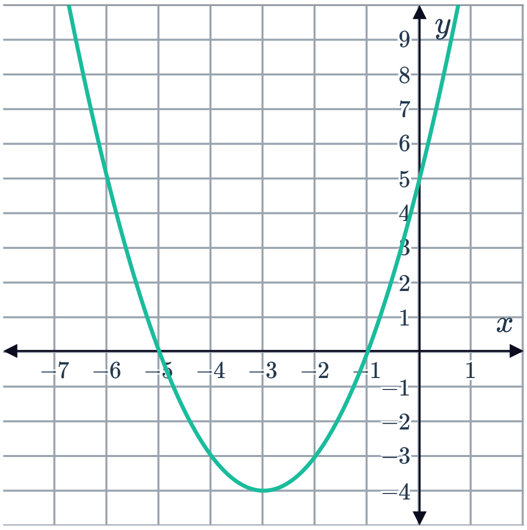
b $a=0$

10

a $y=(x+1)(x+5)$

b $y=5$

c

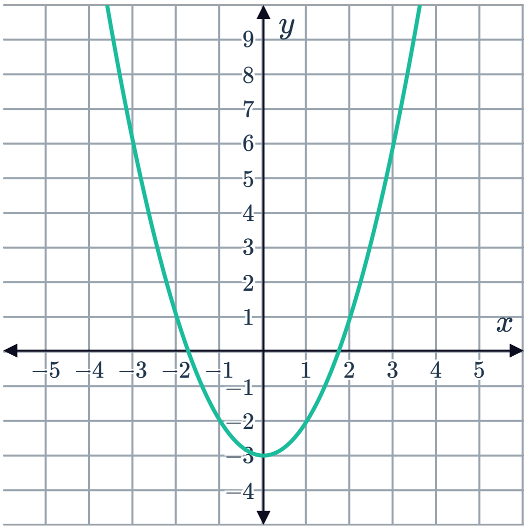


11

a $y=x^2-3$

b -2

c



12

a infinitely many

b infinitely many

c 1

d $y = -\frac{3x^2}{25} - \frac{6x}{25} + \frac{72}{25}$

13 $y = \frac{x^2}{4} + 6x + 39$

14 $b = 4$

15 $c = 4$

16 $k = \frac{1}{3}$

17

a $a + b + c = 3, 9a + 3b + c = -1, 16a + 4b + c = 0$

b $c = 8$

c $a = 1$

d $b = -6$

e $x^2 - 6x + 8$

Maximum and minimum values

18

a Maximum

b $(0, -4)$

19

a Concave up

b $x = 9$

c $y = -71$

20 $y = 0$

21

a -0.087

b -1.24

22

a $x = -6$

b $m = 330$

23

a $x = -10$

b $m = -697$

Graphing quadratics using technology

24 $y = (x-2)^2 + 3$

25

a Concave up b $x = 1, -5$ c $y = -5$ d $x = -2$

e $(-2, -9)$

26 $-\frac{81}{4}$

27

a $x = -3.1$ b -16.61

28

a Concave up b $y = 9$ c $x = -3$ d $x < -3$

29

a Concave down b $y = -6$ c $x = -3$ d $x < -3$

30

a $(-6, -9)$ b $(0, 135)$

Applications

31

a Negative a, positive c b $c = 2560$

c $a = -\frac{1}{10}$ d $\frac{25\,071}{10}$ m

32

a $y = -\frac{1}{20}(x-18)^2 + 16.2$ b 4071.50 m^2

33

a $A(0, 2.5), B(2, 0), C(6, 0)$ b 2

c $4a + k = \frac{5}{2}$ d $0 = 16a + k$

e 5 f 10

34

a $A(0,3.5), B(3,0), C(7,0)$

b 3

c $9a+k=\frac{7}{2}$

d $0=16a+k$

e $a=-\frac{1}{2}$

f $k=8$

35

a $(180,98)$

b $y=-\frac{49}{16\,200}(x-180)^2+98$

c Below

36

a $A(18,0), B(0,108)$

b $y=-\frac{1}{3}(x-18)(x+18)$

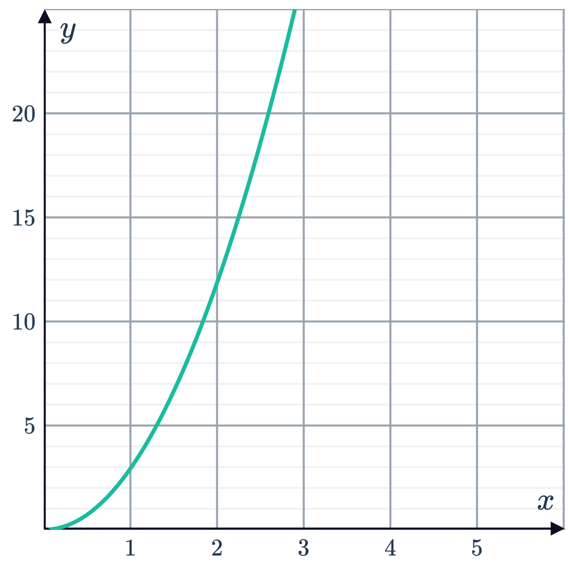
c $x=9\sqrt{2}$

37

a

x	0	1	2	3	4
y	0	3	12	27	48

b



c 0 metres

d 0

e $(0,0)$

38

a Initially, the object has not fallen any distance and is $700\ \mathrm{m}$ above the ground.

b decreasing

c

i True

ii True

iii False

39

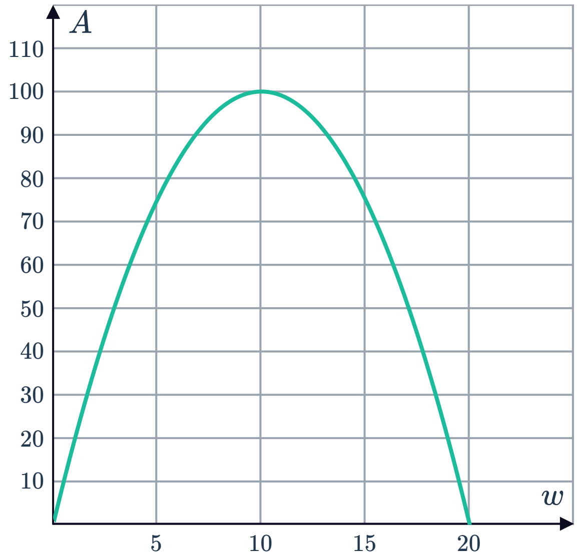
a After 4.5 seconds, the pebble is 40.5 metres above the ground. The highest distance the pebble reaches above the ground is 40.5 metres.

b $t = 0, 9$

c The times when the pebble is on the ground.

40

a



b $w = 0, 20$

c There is no corresponding rectangle because its area would be 0.

d 100

41

a $0, 40$

b 20

c The largest rectangular area that can be fenced off with 80 m of wire.

42

a $s = 12$

b 337

43

a $t = 20$

b 100 metres

44

a $13 - x$

b $x(13 - x)$

c 6.5

d 42.25 m^2

Applications using graphing calculators

45

a $(0,0)$

b The initial distance fallen by the object.

46 $-14, 14\ \mathrm{m/s}$

47

a $176.4\ \mathrm{m}$

b $6\ \mathrm{s}$

c $12\ \mathrm{s}$

48

a $900\ \mathrm{m}$

b Initially, the object has not fallen any distance.

c $13.55\ \mathrm{s}$

d After 13.55 seconds the object is on the ground.

49

a $0, 30$

b 15

c The largest rectangular area that can be fenced off with $60\ \mathrm{m}$ of wire.

50

a $4\ \mathrm{s}$

b $10\ \mathrm{s}$